Structural Selection of Effective Constants

From MAAT Basins to MaxEnt-Weighted RG Bridge Tests

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Abstract

We investigate whether effective physical constants can be interpreted as occupying structurally preferred regions of parameter space. Starting from a sequence of phenomenological MAAT-based benchmarks, we first construct structural scores for dimensionless constants such as the electromagnetic coupling, a mass-ratio proxy, and a cosmological-constant proxy. These tests produce a stable low-defect basin close to the observed order-of-magnitude region.

We then extend the benchmark to a Standard-Model-like setting by introducing a one-loop renormalisation-group (RG) bridge from ultraviolet parameters to infrared effective observables. The observed Standard Model proxy point is used only as a comparison marker, not as an optimisation target. The resulting structural functional selects a low-defect basin that overlaps the observed proxy region. Finally, a direct-term holdout test removes selected direct score terms and tests whether the remaining coupled sectors still constrain the held-out observable.

We further remove one heuristic assumption by calibrating the sector weights λ_a through a maximum-entropy procedure. In the resulting v12/v13 extension, robustness receives the largest inferred weight, followed by the connectivity and generativity sectors. This provides an effective, data-driven weighting layer while remaining phenomenological.

The results support a cautious conclusion: structural selection may constrain effective constants in addition to RG dynamics. They do not constitute a first-principles derivation of the constants, nor a precision Standard Model fit.

1 Status and Scope

This paper is a phenomenological benchmark study. It does not claim to derive the Standard Model, solve the hierarchy problem, determine the cosmological constant, or prove that the observed constants are uniquely selected.

The central question is narrower:

> *Does a structural-selection functional define low-defect basins in constant space, and do observed effective constants lie in or near such basins?*

The guiding architecture is

$$\text{generative parameters} \longrightarrow \text{effective constants} \longrightarrow \text{structural score} \longrightarrow \text{selection ranking}.$$

2 Observed Values and Sources

The comparison values used in the benchmark are not fit targets. They enter only after optimisation, as reference markers in the plots and tables.

For low-energy constants such as the fine-structure constant and mass ratios, the numerical reference values follow CODATA/NIST recommended values [1, 2]. For particle-physics inputs

such as $\alpha_s(M_Z)$, weak mixing, Higgs and top-sector quantities, we use PDG-style values at or near the electroweak scale [3, 4]. Because the present work is not a precision fit, the values are used as representative comparison proxies rather than uncertainty-weighted likelihood inputs.

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3 Structural Functional

For each structural sector a, we define a non-negative defect d_a and a support

$$\Gamma_a = \frac{1}{1+d_a}. \tag{1}$$

The structural score is

$$F_{\mathrm{MAAT}} = -\sum_a \lambda_a \log(\epsilon + \Gamma_a). \tag{2}$$

The v1–v11 benchmarks use the equal-weight baseline $\lambda_a = 1$. In v12/v13, this heuristic choice is replaced by a maximum-entropy calibration of the effective sector weights.

The five sectors are interpreted as:

- H: equation, gauge, or electroweak consistency;
- B: balance of mass, Yukawa, and Higgs-sector scales;
- S: generative capacity for atomic or chemical structure;
- V: interaction coherence;
- R: robustness, stability, and cosmological-constant suppression.

The stability score is computed as

$$\mathrm{Stability} = \min\left[R,\ (HBSV)^{1/4}\right]. \tag{3}$$

3.1 Maximum-Entropy Sector Weights

To reduce the arbitrariness of the sector weights, the v12 benchmark calibrates λ_a as Lagrange multipliers of a maximum-entropy distribution over an ensemble of defect vectors. For a discrete ensemble X_i with sector defects $d_a(X_i)$, define

$$P_\lambda(X_i) = \frac{\exp[-\sum_a \lambda_a d_a(X_i)]}{Z(\lambda)}, \qquad Z(\lambda) = \sum_i \exp\left[-\sum_a \lambda_a d_a(X_i)\right]. \tag{4}$$

The calibration imposes target mean defects

$$\mathbb{E}_\lambda[d_a] \approx d_a^* \tag{5}$$

and optimises the positive weights using $\lambda_a = \exp(\ell_a)$. In v12 the target means are chosen as neutral ensemble medians of a synthetic defect ensemble. Therefore the resulting weights should

be interpreted as effective MaxEnt-calibrated benchmark weights, not as unique microscopic constants.

The v13 bridge uses the normalised weighted score

$$F_{\mathrm{MAAT}}^{(\lambda)} = -\frac{1}{\Lambda_\Sigma} \sum_a \lambda_a \log\left(\frac{\epsilon + \Gamma_a}{\epsilon + 1}\right), \qquad \Lambda_\Sigma = \sum_a \lambda_a, \tag{6}$$

so that the weighted score remains comparable in scale to the equal-weight baseline.

Table 1: v12 maximum-entropy sector-weight calibration. The expected defects match the target means to within about 10^{-3} relative accuracy.

Sector	λ_a	Share	Target d_a^*	Expected $\mathbb{E}_\lambda[d_a]$
H	1.610873	0.0728	0.147084	0.147115
B	2.428297	0.1097	0.140323	0.140389
S	4.623655	0.2088	0.068124	0.068168
V	4.644912	0.2098	0.065946	0.065998
R	8.832344	0.3989	0.030535	0.030560

The inferred ordering

$$R > V \approx S > B > H$$

is structurally meaningful: the robustness sector receives the strongest effective weight, while connectivity and generativity form the next dominant pair. This is consistent with the interpretation that stable effective constants are selected less by direct numerical proximity alone than by robustness and cross-sector compatibility.

4 Development Path: v1–v13

The natural-constants benchmark was developed as a sequence of increasingly structured tests. Table 2 summarises the progression.

Table 2: Development path of the natural-constants and Standard-Model bridge tests.

Version	File	Purpose
v1	`naturkonstante.py`	Initial direct structural optimisation.
v2	`naturkonstante_v2_structure_scan.py`	Log-space structure scan.
v3	`naturkonstante_v3_physics_constraints.py`	Physics viability bands.
v4	`naturkonstante_v4_stellar_chemistry.py`	Chemistry/fusion proxies.
v5	`naturkonstante_v5_robustness_scan.py`	Multi-seed robustness.
v6	`naturkonstante_v6_ablation_scan.py`	Sector ablation tests.
v7	`naturkonstante_v7_landscape_heatmap.py`	Basin visualisation.
v8	`naturkonstante_v8_gradient_flow.py`	Gradient-flow attractor test.
v9	`naturkonstante_v9_rg_maat.py`	Toy RG plus MAAT score.
v10	`naturkonstante_v10_multiseed.py`	RG multi-seed robustness.
v11	`standard_model_rg_maat_v11_holdout.py`	SM-like holdout test.
v12	`naturkonstante_v12_maxent_lambda.py`	MaxEnt calibration of sector weights.
v13	`naturkonstante_v13_maxent_sm_bridge.py`	MaxEnt-weighted constants bridge.

5 Natural-Constants Basin

The first benchmark treats the effective constants

$$(\alpha, \mu, \Lambda)$$

as dimensionless parameters, where α is an electromagnetic coupling, μ is a mass-ratio proxy, and Λ is a cosmological-constant proxy in Planck units. The score is built from broad structural bands for atomic structure, mass hierarchy, chemistry, fusion-like energy generation, and cosmological robustness.

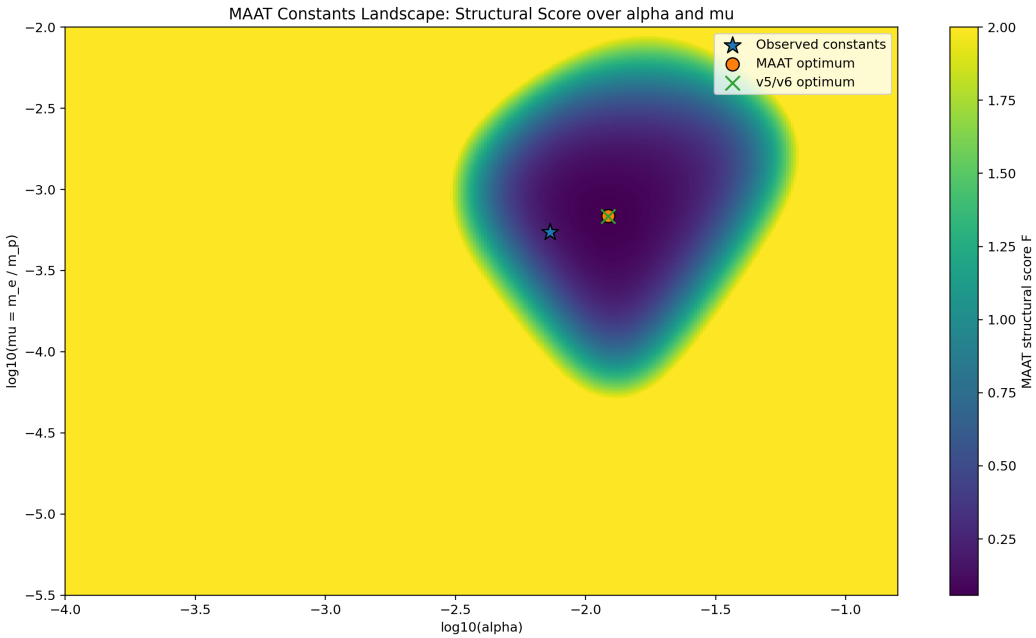

Figure 1: Structural landscape over (α, μ) at fixed cosmological-constant proxy. The observed comparison point and the MAAT optimum lie in the same connected low-defect basin. This indicates basin-level proximity, not exact prediction.

The v8 gradient-flow test probes whether the low-defect basin acts as an attractor under structural descent. Multiple initial conditions flow into the same region, indicating that the basin is not a single-point numerical accident.

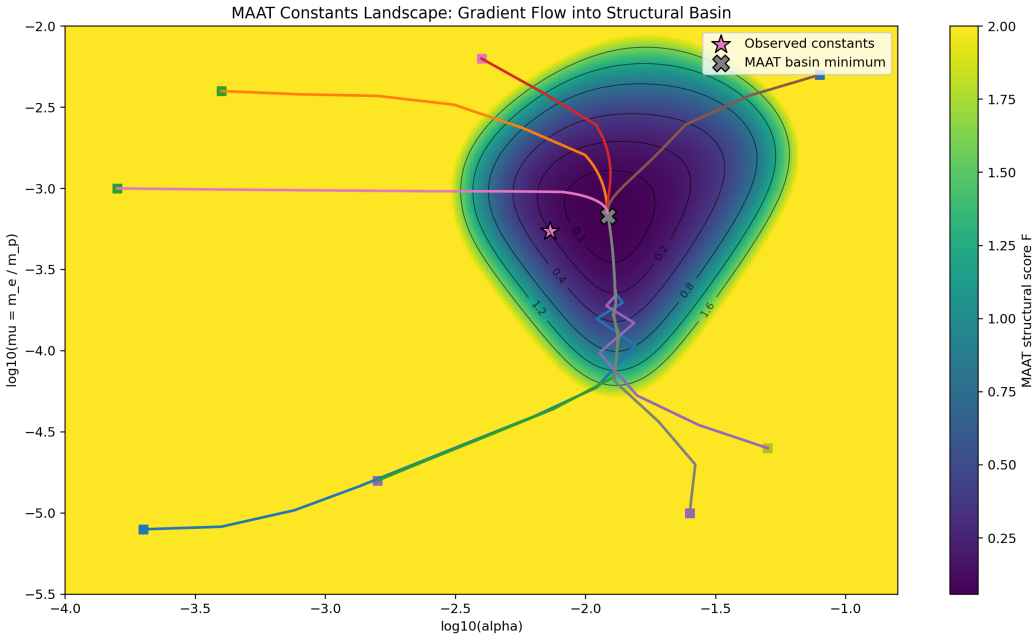

Figure 2: Gradient-flow trajectories in the structural landscape. The trajectories converge toward the same low-defect basin, supporting the interpretation of the selected region as a structural attractor.

The v10 multi-seed RG benchmark finds the stable region

$$\alpha_{\mathrm{IR}} \approx 1.22 \times 10^{-2}, \qquad \mu \approx 6.7 \times 10^{-4}, \qquad \Lambda \approx 2.4 \times 10^{-129}.$$

The first two values lie near the observed order-of-magnitude region, while the cosmological-constant proxy remains substantially displaced. This is evidence for a structural basin, not a derivation of the constants.

6 Standard-Model RG Bridge

The next step replaces the direct constant scan by a Standard-Model-like RG bridge. The ultraviolet parameter vector is

$$\theta_{\mathrm{UV}} = (g_1, g_2, g_3, y_t, y_e, \lambda_H, \Lambda_{\mathrm{CC}}) \,, \tag{7}$$

with g_1 GUT-normalised. These parameters are evolved from

$$\mu_{\mathrm{UV}} = 10^{16}\,\mathrm{GeV}$$

to

$$\mu_{\mathrm{IR}} = M_Z$$

using a minimal one-loop Standard-Model-like RG system. The infrared proxy observables are

$$\theta_{\mathrm{IR}} = \left(\alpha_{\mathrm{em}}, \alpha_s, \sin^2\theta_W, y_e, y_t, \lambda_H, \Lambda_{\mathrm{CC}}\right). \tag{8}$$

The benchmark does not perform a precision Standard Model fit. Threshold corrections, full two-loop running, uncertainty propagation, and vacuum metastability calculations are left for future work.

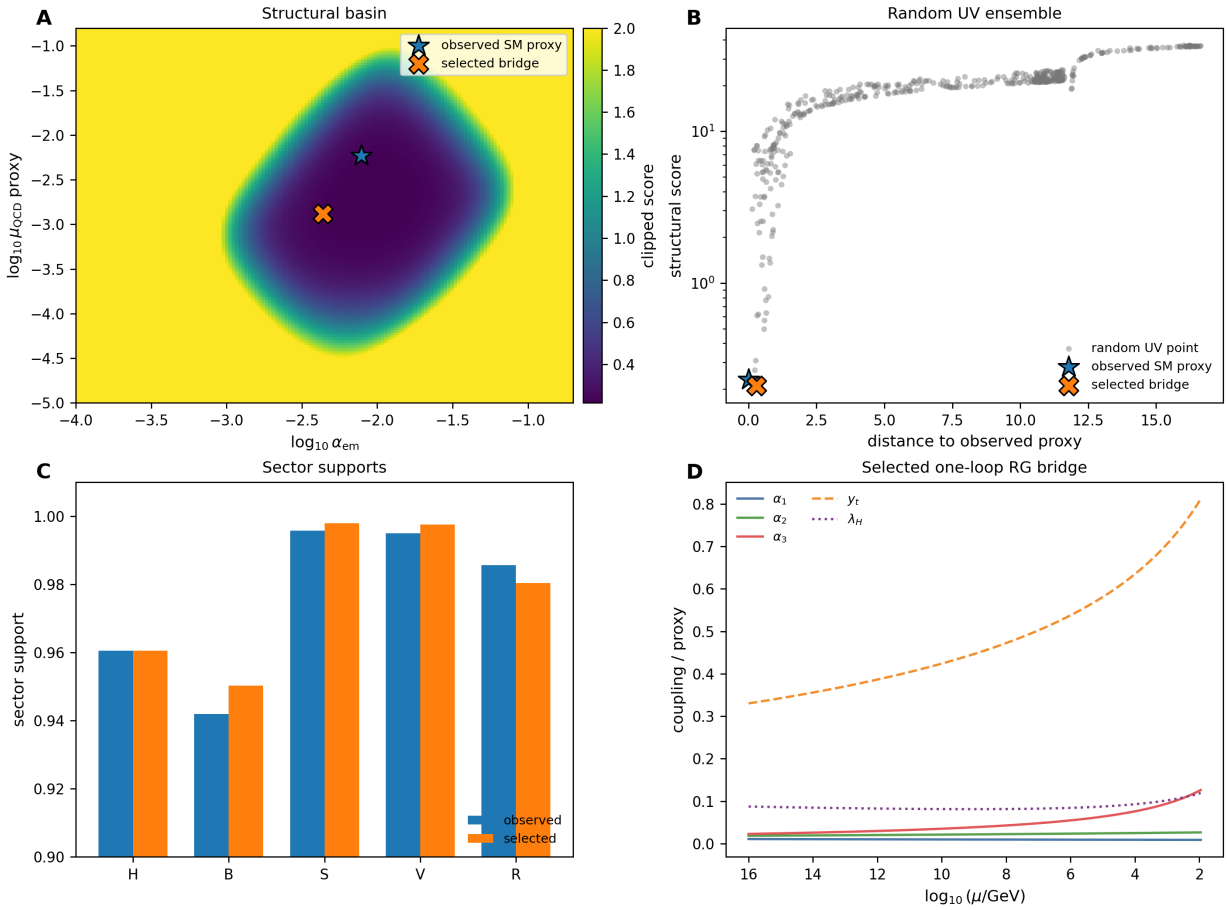

Figure 3: Standard-Model RG bridge benchmark. (A) Structural basin in the effective $(\alpha_{\rm em}, \mu_{\rm QCD})$ plane. (B) Random UV ensemble: points farther from the observed proxy generally receive larger structural scores. (C) Sector supports for the observed proxy and selected bridge point. (D) One-loop RG bridge for the selected point.

The observed Standard Model proxy score and selected bridge score are:

$$F_{\rm obs} = 0.231789, \tag{9}$$

$$F_{\rm bridge} = 0.210446, \tag{10}$$

$$F_{\rm bridge} - F_{\rm obs} = -0.021342. \tag{11}$$

The selected point is slightly lower in structural score, but the observed proxy lies in the same low-defect basin. This is the scientifically relevant result: the score identifies a region compatible with observed effective structure, not a unique observed point.

Table 3: Observed Standard Model proxy compared with the selected bridge point.

Observable	Observed SM proxy	Selected bridge point
$\alpha_{\rm em}$	0.007815	0.004365
α_s	0.117900	0.126098
$\sin^2\theta_W$	0.231220	0.165686
y_t	0.990904	0.811444
λ_H	0.129281	0.119000

7 v11 Holdout Test

The v11 benchmark tests cross-sector predictivity. A direct-term holdout removes one direct band condition from the score while leaving indirect appearances through coupled structural proxies active. For example, in the α_{em} holdout, the direct electromagnetic band is removed, but α_{em} still enters through chemistry and fusion proxies.

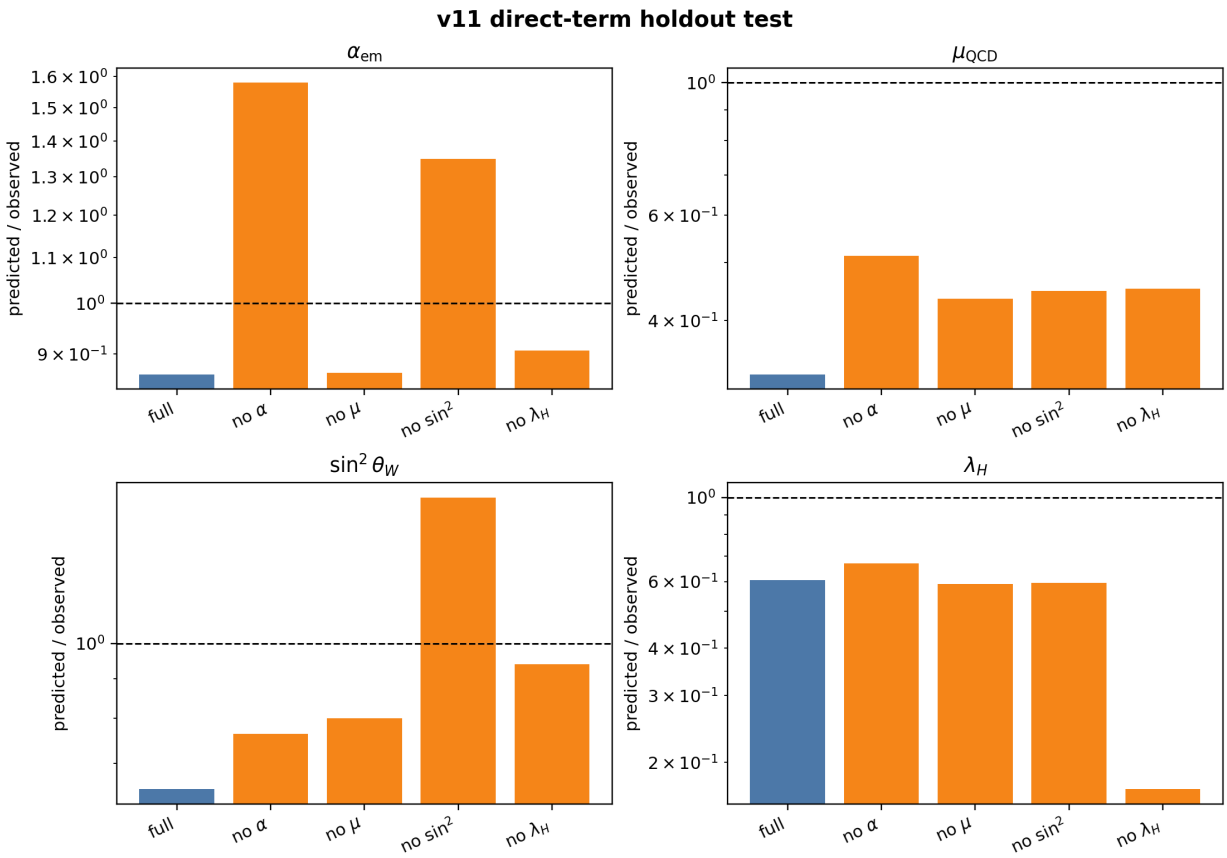

Figure 4: v11 direct-term holdout test. Each panel shows predicted/observed ratios across the full score and four holdout cases. Values near unity indicate that the remaining sectors still constrain the held-out quantity to the observed order-of-magnitude region.

Table 4: v11 direct-term holdout results.

Case	Held-out observable	Predicted/observed	$\lvert\Delta\log_{10}\rvert$	Stability
Full score	none	0.863 (α_{em})	0.064	0.9785
No direct α_{em}	α_{em}	1.581	0.199	0.9750
No direct μ_{QCD}	μ_{QCD}	0.435	0.362	0.9780
No direct $\sin^2\theta_W$	$\sin^2\theta_W$	1.543	0.188	0.9833
No direct λ_H	λ_H	0.170	0.770	0.9744

The result is partially successful. The α_{em} and $\sin^2\theta_W$ holdouts remain within a factor of about 1.6 of the observed proxies. The μ_{QCD} holdout remains within a factor of about 2.3. The Higgs quartic holdout is weaker, giving about 17% of the observed proxy value. This identifies the Higgs sector as the main weakness of the present toy bridge.

8 v13 MaxEnt-Weighted Constants Bridge

The v13 benchmark applies the MaxEnt-calibrated weights from Table 1 to a simplified constants bridge. The generative variables are an ultraviolet electromagnetic coupling, an RG-span proxy, an electron-Yukawa proxy, a QCD-scale proxy, and a cosmological-constant proxy. These are mapped to effective infrared quantities $(\alpha_{\mathrm{IR}}, \mu_{\mathrm{eff}}, \Lambda)$ and evaluated using the weighted structural score in Eq. (6).

Table 5: v13 MaxEnt-weighted constants bridge. The first two constants remain near the observed order-of-magnitude region, while the cosmological-constant proxy remains strongly displaced.

Quantity	v13 selected value	Reference proxy	Selected/reference
α_{IR}	1.2277×10^{-2}	7.2974×10^{-3}	1.682
μ_{eff}	6.7492×10^{-4}	5.4462×10^{-4}	1.239
Λ	2.3841×10^{-129}	10^{-122}	2.38×10^{-7}

The MaxEnt-weighted run gives

$$F^{(\lambda)}_{\mathrm{MAAT}} = 0.003387, \qquad \text{Stability} = 0.993605, \qquad F_{\mathrm{total}} = 0.041234.$$

The corresponding sector supports are

$$(H, B, S, V, R) = (0.989878,\ 0.993827,\ 0.994631,\ 0.996095,\ 0.999944).$$

Thus the MaxEnt-weighted extension preserves the main qualitative conclusion of the earlier benchmarks: the electromagnetic and mass-ratio proxies are structurally constrained to the observed order-of-magnitude region, while the cosmological-constant proxy remains unresolved.

9 Interpretation

The experiments support the following cautious statement:

> *Effective constants may be constrained by structural-selection criteria in addition to RG dynamics.*

The stronger claim that the constants have been derived is not supported by the current benchmark. The viability bands are phenomenological and the RG system is simplified. The v12/v13 MaxEnt calibration reduces the arbitrariness of the sector weights, but it does not derive them from microscopic physics.

The value of the benchmark lies in its falsifiability. The holdout test can fail for specific sectors, as seen for λ_H. This is scientifically useful: it separates robust cross-sector constraints from sectors requiring additional physical input.

10 Reproducibility

The numerical material for this paper is archived in the public repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments

The natural-constants benchmark sequence is stored in:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/natural_constants_selection

It contains the scripts `naturkonstante.py` through `naturkonstante_v13_maxent_sm_bridge.py`, together with the generated basin and gradient-flow figures used in Figs. 1 and 2, and the JSON outputs for the v12/v13 MaxEnt calibration.

The Standard-Model bridge and v11 holdout tests are stored in:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/standard_model_bridge

The core Standard-Model bridge scripts are:

- `standard_model_rg_maat_bridge.py`
- `standard_model_rg_maat_summary_figure.py`
- `standard_model_rg_maat_v11_holdout.py`

The intended reproducibility loop is:

$$\text{model assumptions} \to \text{generated ensemble or RG bridge} \to \text{effective observables} \\ \to \text{MAAT score} \to \text{JSON results and plots.}$$

The compiled PDF is stored with the paper series in the repository `documentation/` directory.

11 Limitations

- The structural bands are phenomenological and not derived from first principles.
- The RG bridge is one-loop and approximate.
- Threshold corrections and uncertainty propagation are not included.
- The cosmological-constant sector is a proxy and not a solution to the cosmological constant problem.
- The Higgs-sector holdout is weak, indicating missing physics in the current model.
- The v12 sector weights are MaxEnt-calibrated effective weights, but the underlying defect ensemble and target constraints remain phenomenological.

12 Conclusion

The natural-constants and Standard-Model bridge benchmarks show that MAAT-type structural selection can define stable low-defect basins in effective-constant space. The observed comparison points lie within or near these basins, v11 holdout tests demonstrate partial cross-sector predictivity for gauge and chemistry-related observables, and the v12/v13 MaxEnt extension shows that the equal-weight assumption can be replaced by calibrated effective sector weights.

The appropriate conclusion is not that the constants have been derived, but that structural selection provides a testable additional layer on top of RG dynamics. The next required step is a precision version of the bridge using full Standard Model running, thresholds, uncertainties, and a stricter held-out prediction protocol.

References

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